Scheduling Theory Assignment

SMRTTECH 4SC3
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1.

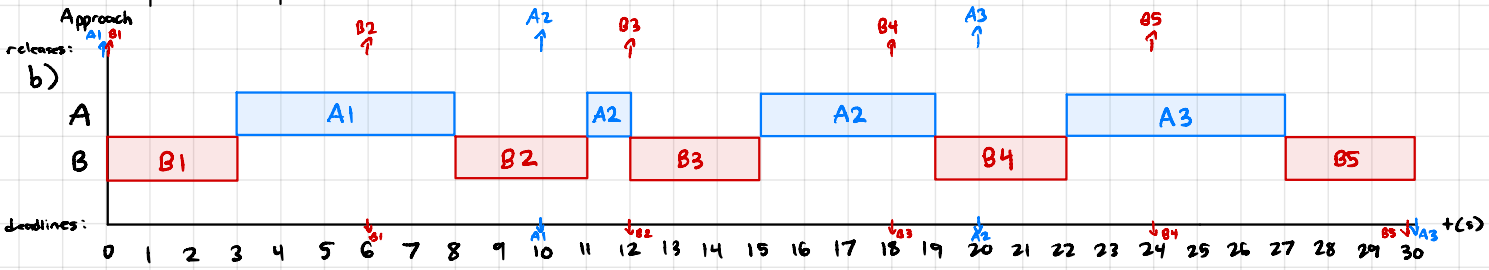
a)

Job	Release (r)	Absolute deadline (d)
A1	0	0 + 10 = 10
A2	10	10 + 10 = 20
A3	20	20 + 10 = 30
B1	0	0 + 6 = 6
B2	6	6 + 6 = 12
B3	12	12 + 6 = 18
B4	18	18 + 6 = 24
B5	24	24 + 6 = 30

$U = \sum_i \frac{C_i}{T_i} = \frac{5}{10} + \frac{3}{6} = 1.0$

$H = LCM(T_i) = LCM(10, 6) = 30\,s$

b)



Job	d	f	$L = f - d$	$R = f - r$
A1	10	8	−2	8
A2	20	19	−1	9
A3	30	27	−3	7
B1	6	3	−3	3
B2	12	11	−1	5
B3	18	15	−3	3
B4	24	22	−2	4
B5	30	30	0	6

$L_{max} = 0$ (Job B5)

Since all L's are ≤ 0, all deadlines were met.

c) $U = \sum_i \frac{C_i}{T_i} = \frac{5}{10} + \frac{3.3}{6} = 1.05 > 1$

$(3)(5) + (5)(3.3) = 31.5\,s$, but only 30s of service time exists. Since $U > 1$, no schedule of any kind can meet this workload indefinitely. The original workload had $U = 1$ (no slack or idle time). The EDF schedule contains 8 approach switches. If every switch between approaches costs a positive clearance time $\varepsilon > 0$, the total demand per hyperperiod becomes $30 + (\#\text{ of switches}) \times \varepsilon$, which exceeds the 30s available. Because there is no idle time to absorb it, some jobs finish late and the backlog accumulates every hyperperiod. Any schedule serving both approaches needs at least some switches, so the original workload becomes infeasible, not just EDF.

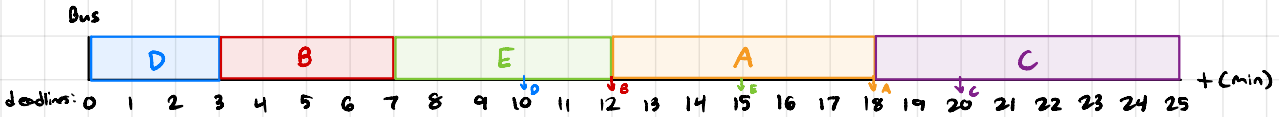
d) These requirements violate two assumptions: jobs can be interrupted and resumed without loss, and switching/clearance time is 0. The EDF trace splits A2 into $[11, 12)$ and $[15, 19)$ and switches approaches 8 times. A real signal needs uninterrupted greens and clearance at each switch. Because $U = 1$ leaves no idle time, any positive clearance pushes work past deadlines, so the trace cannot prove physical feasibility. L_{max} also does not directly measure vehicle waiting: it compares job completion with the abstract deadline $D = T$, ignoring vehicle arrivals, queue formation, and red-phase delay. A vehicle can wait a long time even when $L_{max} \leq 0$.

2.

a) Resource: one shuttle (a single machine), serving one service block at a time
Release pattern: all 5 requests are released at $t = 0$
Preemption rule: none. Each block is served without interruption.
EDD minimizes Lmax because with a single resource, common release times, and nonpreemptive jobs of fixed duration, an exchange argument applies. Suppose a schedule has job j immediately before job i while $d_i < d_j$. Swapping them lets i finish earlier and j finish at the same time the pair previously ended. The swap does not increase the maximum lateness of the pair: i's lateness drops, and j now finishes at the old finish time of i, which is no later than before relative to its later deadline. Repeating swaps gives the earliest-due-date order without ever increasing Lmax, so EDD is optimal for Lmax.

EDD minimizes the worst lateness. This is not the same as guaranteeing every target is met. If the minimum Lmax is ≤ 0, all targets are met (and EDD finds a feasible schedule whenever one exists). If the minimum Lmax is > 0, EDD still gives the best possible worst case, but some target must be missed.

b) EDD order: D, B, E, A, C



Request	C	d	s	f	L = f − d	Tardiness
D	3	10	0	3	−7	0
B	4	12	3	7	−5	0
E	5	15	7	12	−3	0
A	6	18	12	18	0	0
C	7	20	18	25	+5	5

Lmax = 5 (request C).
One L is > 0. Not all targets are met. C completes 5 minutes late.

c) Fixed order: A, B, C, D, E, all $r = 0$

Request	s	f	d	L = f − d	W = s − r
A	0	6	18	−12	0
B	6	10	12	−2	6
C	10	17	20	−3	10
D	17	20	10	+10	17
E	20	25	15	+10	20

Fixed order:
Lmax = 10
mean $W = \frac{0 + 6 + 10 + 17 + 20}{5} = 10.6$ min

EDD:
Lmax = 5
mean $W = \frac{0 + 3 + 7 + 12 + 18}{5} = 8$ min

EDD is better on both measures here. However, EDD does not generally guarantee minimum mean waiting. Minimizing mean waiting is achieved by shortest-processing-time-first (SPT), which ignores due times. In this example the durations 3, 4, 5, 6, 7 happen to be in the same order as the due times. In this example the durations 3, 4, 5, 6, 7 happen to be in the same order as the due times, so EDD and SPT coincide and 8 min is also the minimum possible mean. That is a coincidence of this data, not a general property. This is a request-level mean, not passenger-weighted.

d) Total work = 6 + 4 + 7 + 3 + 5 = 25 min. Whatever the order, the last block finishes at $t = 25$, but the latest due time is 20 (for C), so the last request is 5 min late. EDD minimizes Lmax and gives Lmax = 5, which is > 0, so no order meets all original due times.

A common extension δ shifts every lateness down by δ, and the best achievable Lmax with the new due times is $5 - \delta$. For feasibility we need $5 - \delta \leq 0$, so the smallest value is $\delta = 5$ min. New due times: D 15, B 17, E 20, A 23, C 25. The EDD order is unchanged (D, B, E, A, C), finishes are 3, 7, 12, 18, 25, and the lateness values are −12, −10, −8, −5, 0. The recheck gives Lmax = 0, and all targets are met.

c) 1. all jobs are released at the same time ($r = 0$)

2. one resource serves jobs nonpreemptively with fixed durations that do not depend on the order of service

The shuttle satisfies both: all 5 requests are ready at $t = 0$, one bus serves one block at a time, and each duration already includes boarding, travel, and return to dispatch, so no sequence-dependent travel exists. A realistic change is requests arriving at different times, or travel time depending on the previous drop-off. The exchange argument would then no longer hold, since an urgent late arrival or an order-dependent duration can change the optimum, so the theorem must be rechecked.

3.

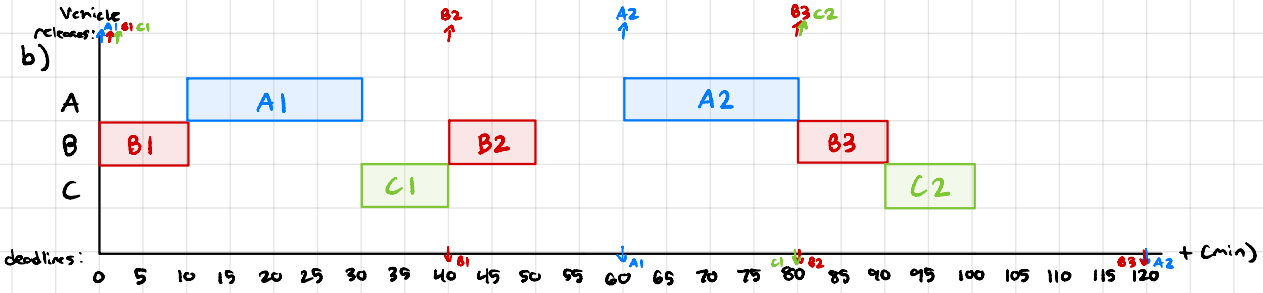
a) RM priority order: B, A, C

$U = \sum_i \frac{C_i}{T_i} = \frac{20}{60} + \frac{10}{40} + \frac{10}{80} = 0.7083$

Bound $B = 3(2^{\frac{1}{3}} - 1) = 0.7798$

Since $U = 0.7083 \leq 0.7798$, the sufficient condition holds. Under the given assumptions (independent periodic jobs, $D = T$, preemptive fixed-priority scheduling, zero switching time, no overhead), RM can meet all deadlines for this workload. The test is sufficient but not necessary.

b)



Jobs with deadlines at or before 120:

Job	d	f	L = f − d
B1	40	10	−30
A1	60	30	−30
C1	80	40	−40
B2	80	50	−30
A2	120	80	−40
B3	120	90	−30

C2 has $d = 160 > 120$, so it is not listed. All deadlines are met. There are no preemptions in this trace. In general, RM is a preemptive fixed-priority policy: a higher priority release (such as B) preempts a lower-priority job. This trace simply has no case where that happens.

c) $h = GCD(60, 40, 80) = 20$ min

$H = LCM(60, 40, 80) = 240$ min

$\frac{H}{h} = \frac{240}{20} = 12$ frames

Frame	Interval	Jobs	Busy	Idle
0	[0, 20)	B1 [0, 10), C1 [10, 20)	20	0
1	[20, 40)	A1 [20, 40)	20	0
2	[40, 60)	B2 [40, 50)	10	10
3	[60, 80)	A2 [60, 80)	20	0
4	[80, 100)	B3 [80, 90), C2 [90, 100)	20	0
5	[100, 120)	idle	0	20
6	[120, 140)	A3 [120, 140)	20	0
7	[140, 160)	B4 [140, 150)	10	10
8	[160, 180)	B5 [160, 170), C3 [170, 180)	20	0
9	[180, 200)	A4 [180, 200)	20	0
10	[200, 220)	B6 [200, 210)	10	10
11	[220, 240)	idle	0	20

Durations: no frame exceeds 20 min and no job crosses a frame boundary. Total busy = 170 min = (U)(240) = (0.7083)(240). Total idle time = 70 min

Releases: each job starts at or after its release (all start at or after its release frame boundary)

Deadlines: all are met.

Repeatability: all 3 tasks release simultaneously at $t = 240$, and no work carries over past the frame ending at 240, so the same table repeats every 240 min.

d) $U = 0.81$ exceeds the RM bound of 0.7798. The sufficient bound therefore cannot confirm schedulability. The test is inconclusive, not a proof of failure. It does not show that RM will miss a deadline. A workload above the bound may or may not be schedulable, so an exact analysis is required (response-time analysis or simulation over the hyperperiod). What can be said is that $U = 0.81 \leq 1$ satisfies the necessary condition, so the workload is not ruled out under the ideal model.

Two assumptions whose failure requires re-analysis:

1. zero switching analysis: if swapping plugs or energizing a different plug takes time, that time is added demand with no charging. This effectively raises U and can push a job past its deadline, especially at $U = 0.81$ where slack is small
2. pause and resume without loss (preemption): if a charging job cannot be paused, a long low-priority job blocks a higher-priority one for its full duration. This blocking delays the high-priority job beyond what the RM analysis assumes and can cause a missed deadline even with U below the bound.